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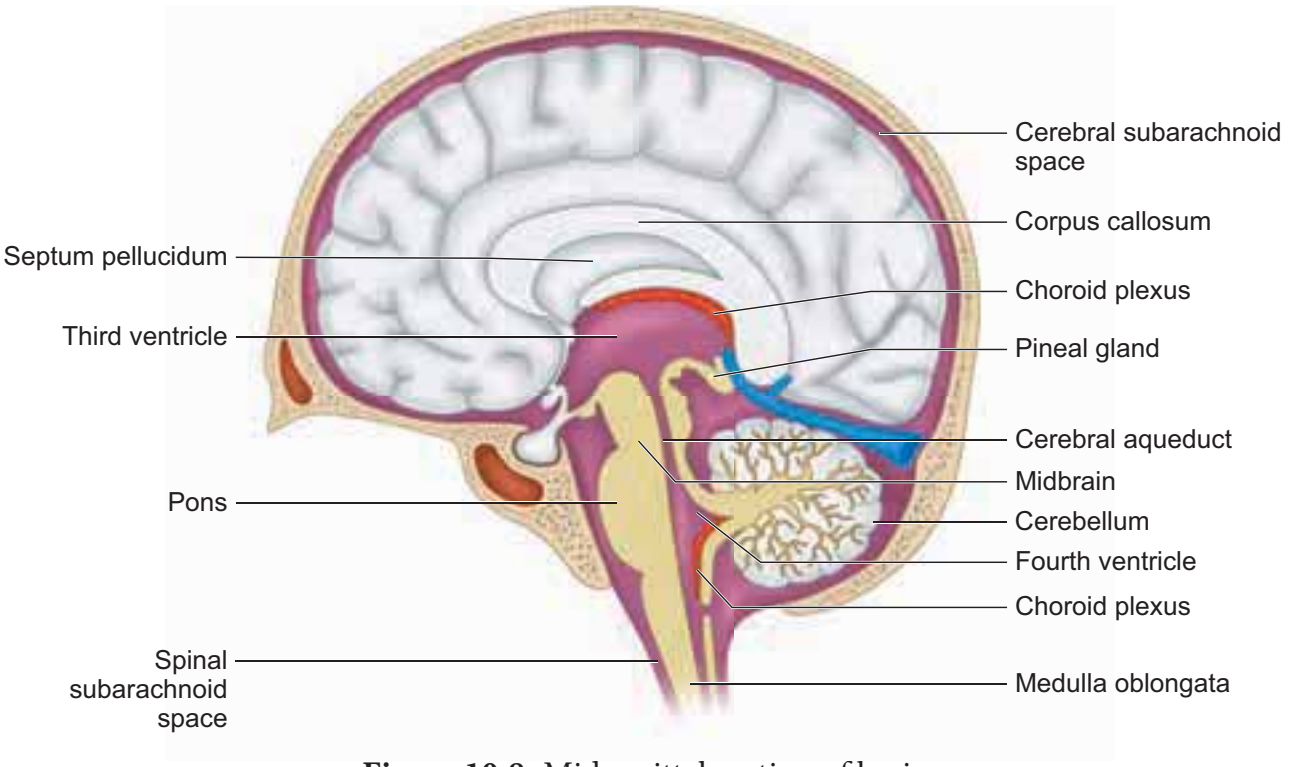


**Figure 10.8** Mid sagittal section of brain

hemispheres are connected by a tract of nerve fibres called **corpus callosum**. Cerebral cortex has three functional areas namely **sensory areas** occur in the parietal, temporal and occipital lobes of the cortex. They receive and interpret the sensory impulses. **Motor area** of the cortex which controls voluntary muscular movements lies in the posterior part of the frontal lobes. The areas other than sensory and motor areas are called **Association areas** that deal with integrative functions such as memory, communications, learning and reasoning. Inner to the cortex is **medulla** which is white in colour and acts as a nerve tract between the cortex and the diencephalon.

**Diencephalon** consists largely of following three paired structures.

**Epithalamus** forms the roof of the diencephalon and it is a non-nervous tissue. The anterior part of epithalamus is vascular and folded to form the anterior **choroid plexus.** Just behind the choroid plexus, the epithalamus forms a short stalk which ends in a rounded body called **pineal body** which secretes the hormone, **melatonin** which regulates sleep and wake cycle.

**Thalamus** is composed of grey mater which serves as a relay centre for

**Table 10.2** Functions of brain lobes

| Structure | Functions |
|---|---|
| Frontal | Behaviour, Intelligence, Memory, Movement |
| Parietal | Language, Reading, Sensation |
| Temporal | Speech, Hearing, Memory |
| Occipital | Visual processing |